LF AI & DATA

The EGERIA Metadata Show

Egeria Webinar

USING INTEGRATION CONNECTORS

Mandy Chessell CBE FREng
Egeria Open Source Project Lead

Date	time	Title	Description	Presenter
January 2022	n/a	n/a	There is no webinar this month. The team is working on creating a new set of monthly Dojo sessions. The first of these sessions will be on the 17th of January 2022 10AM - 17:00PM (UTC) and will be on **How to setup and run Egeria in different environments**. This will include the Kubenetes environment. Zoom Conference https://zoom.us/j/523629111	
7th February 2022	15:00 UTC	**Using an integration connector**	This session covers how to use Integration connectors to connect technologies into Egeria..	**Mandy Chessell**
7th March 2022	15:00 UTC	**How to build an integration connector**	This session covers how to extend Egeria's automated cataloguing to include metadata from a new technology. It describes how automated cataloguing works and the role of the integration connector. It covers the design of the integration connector using examples to illustrate the different approaches and their benefits and and challenges. It shows how to set up a project for a new connector, how to build and package it and finally it shows the new connector running in Egeria.	**Mandy Chessell**
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Coco Pharmaceuticals persona

Jules Keeper, CDO

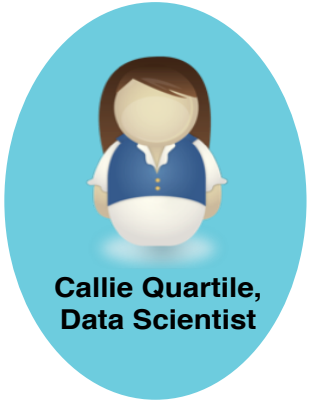
Tessa Tube,
Chief Researcher

Faith Broker
Chief Privacy Officer

Nancy Noah
Cloud Specialist

Erin Overview,
Information Architect

Bob Nitter,
Integration Developer



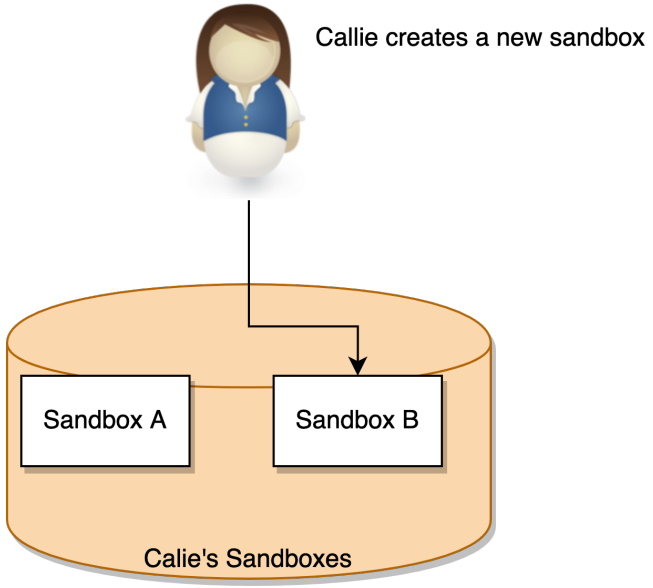
Callie Quartile,
Data Scientist

Gary Geeke
IT Infrastructure

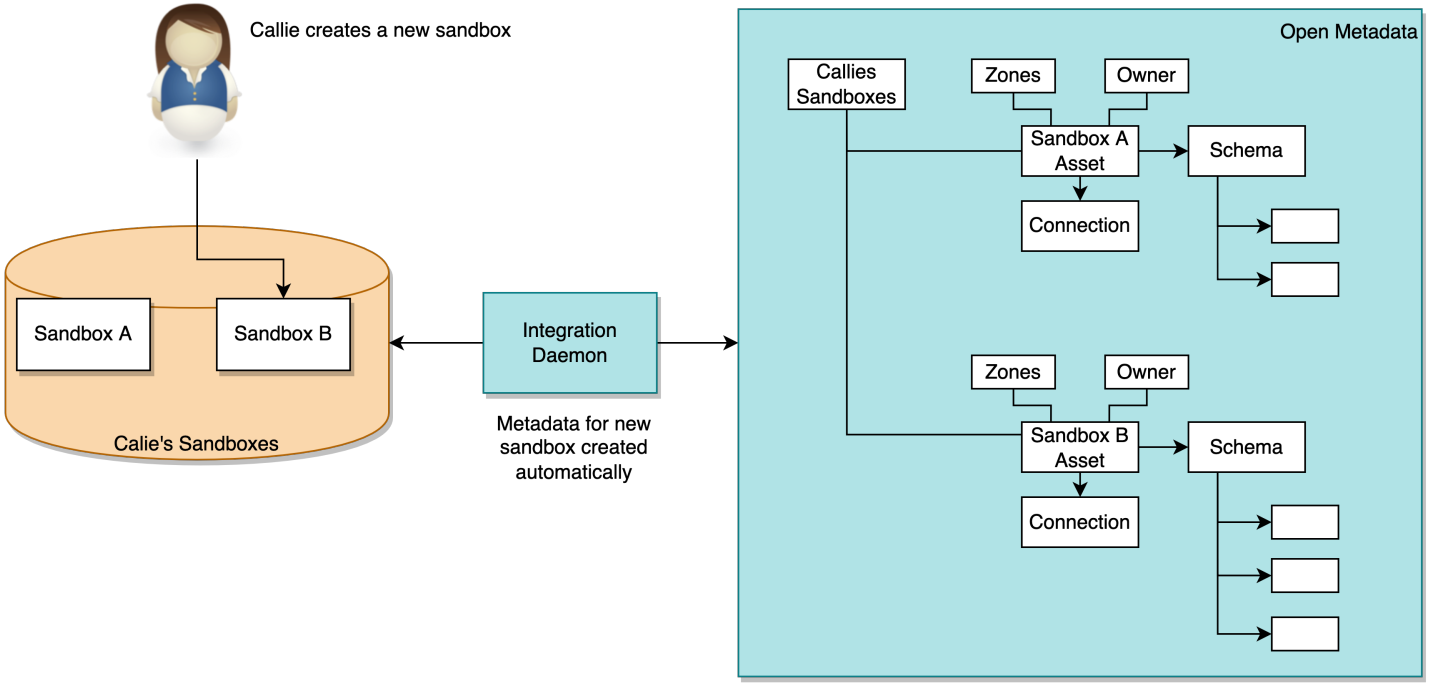
https://odpi.github.io/data-governance/coco-pharmaceuticals/personas/

Integrated cataloguing

- Callie has a database server that she uses to analyze relational data.
- She creates a new sandbox for each type of analysis.
- However, she often forgets to catalog her sandboxes.

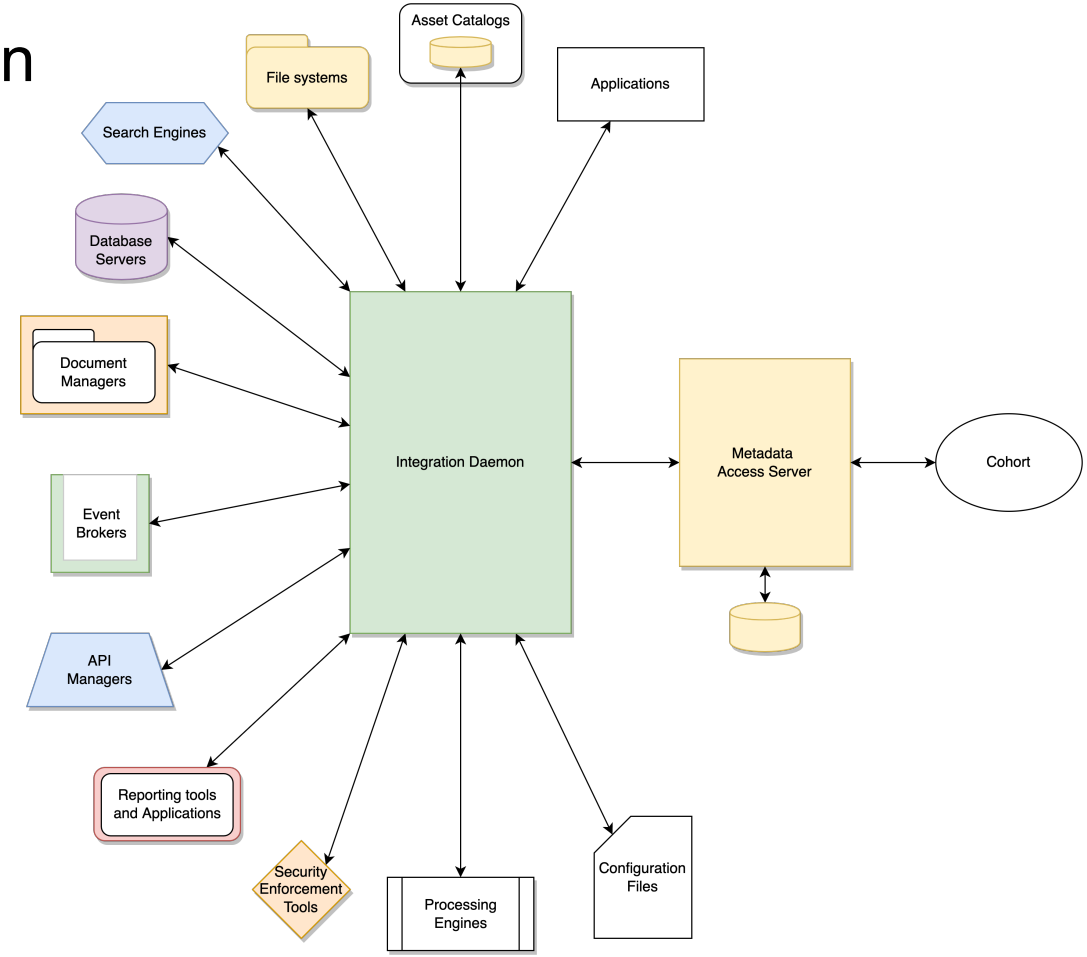


Automated cataloguing of Callie's Sandboxes

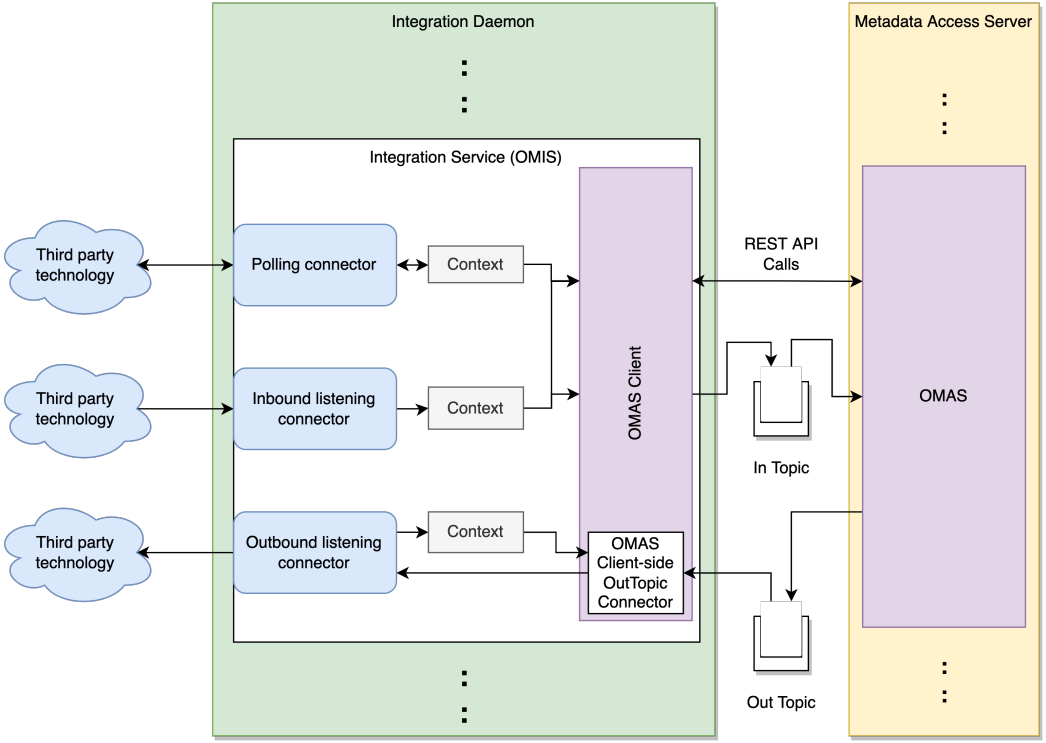


The Integration Daemon

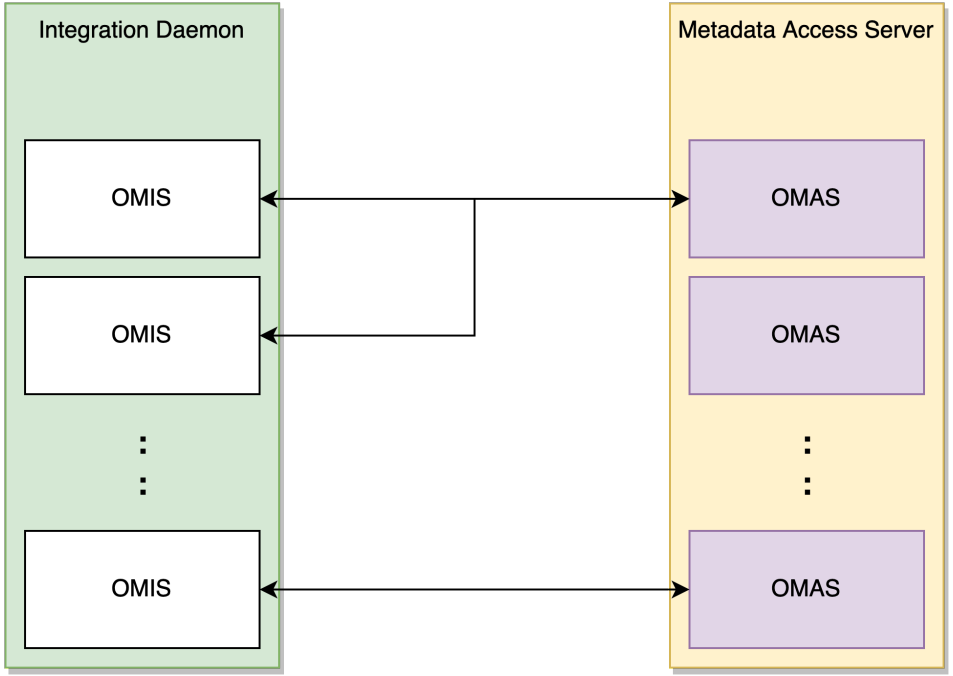
- A type of OMAG Server
- Metadata extraction, capture and delivery



Inside the integration daemon

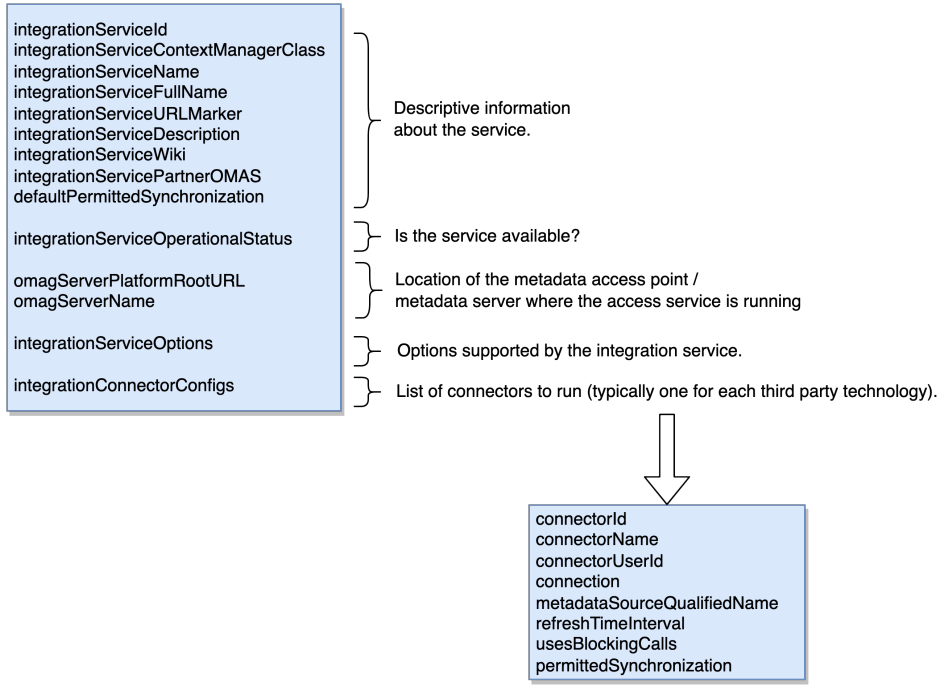


Supporting the metadata needs of different technologies

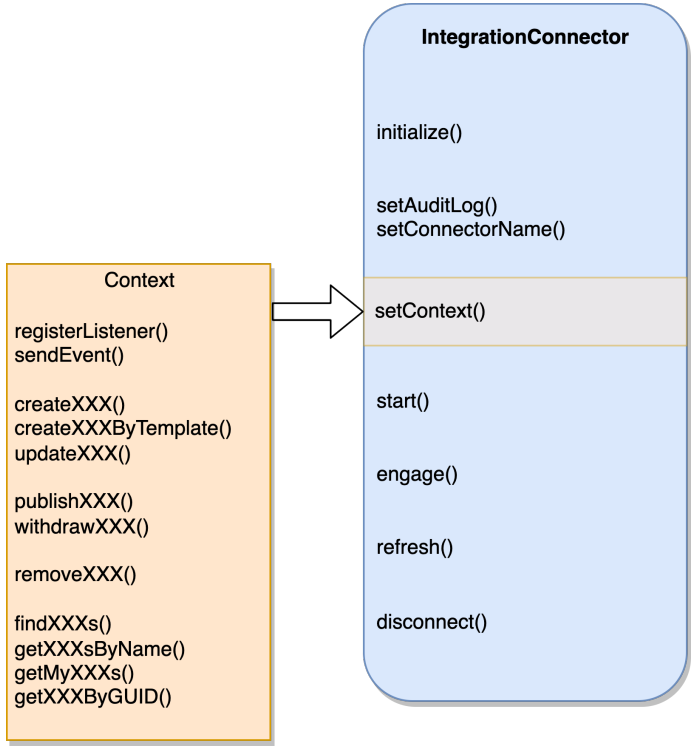


Integration Connector configuration

- The configuration provides the integration daemon with the information it needs to control the lifecycle and runtime support needed by the connector.



Integration Connector Implementation



For saving connector instance identifier and connection object.
Called from the ConnectorBroker.

Provides the logging destination.

Provides the name of the connector for logging.

Set up the integration service specific context.

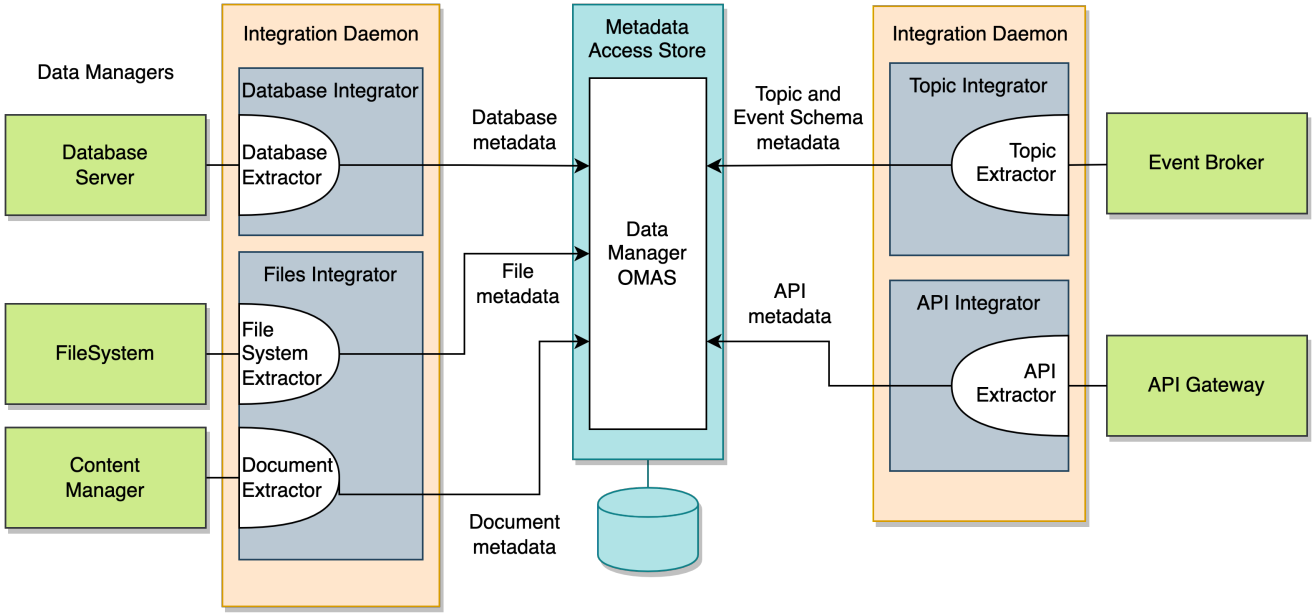
Indicates that the connector is completely configured and can begin processing. This call can be used to register with non-blocking services. For example it can register a listener with the OMAS Out Topic with the context.

Used for blocking calls to wait for new metadata. It is called from its own thread iff the connector is configured to have its own thread. It is recommended that the engage() method returns when each blocking call completes. The integration daemon will pause a second and then call engage() again. This pattern enables the calling thread to detect the shutdown of the integration daemon server.

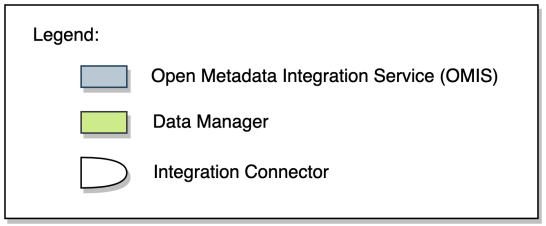
Requests that the connector does a comparison of the metadata in the third party technology and open metadata repositories. Refresh is called when the integration connector first starts and then at intervals defined in the connector's configuration as well as any external REST API calls to explicitly refresh the connector.

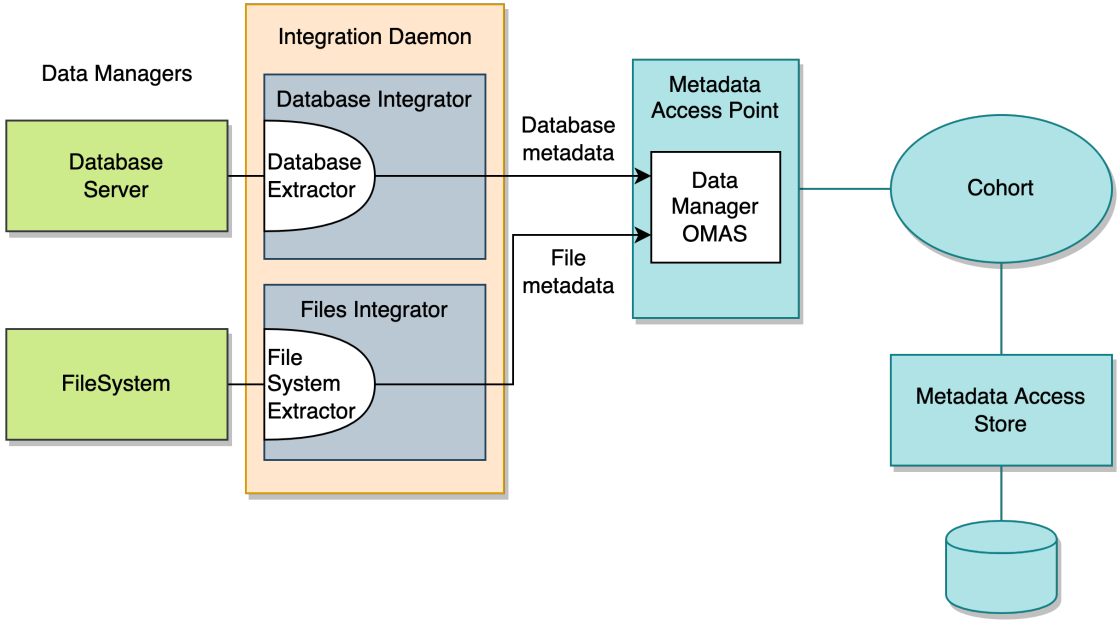
Free up any resources held since the connector is no longer needed.

Metadata Extraction

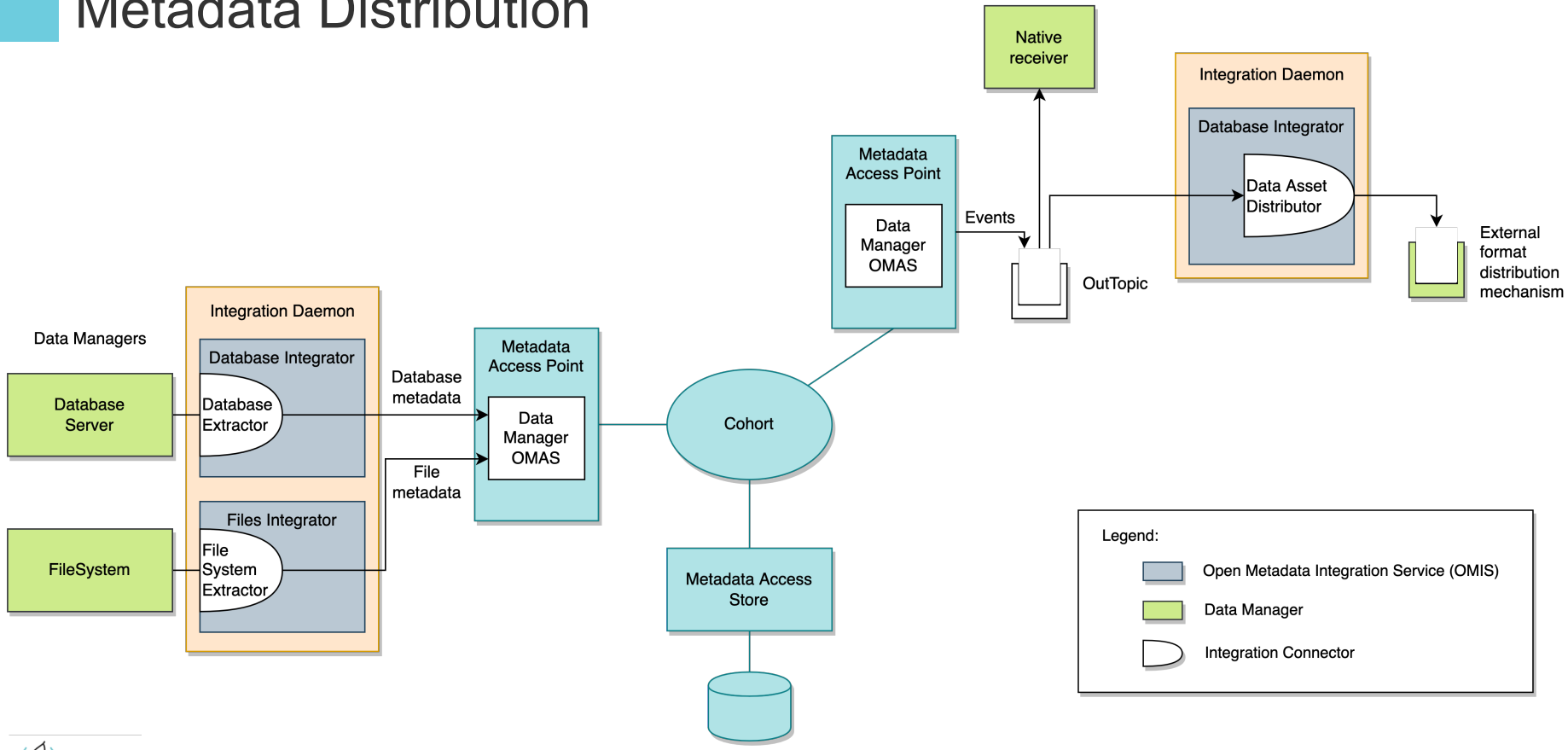


Metadata Extraction (remote store)

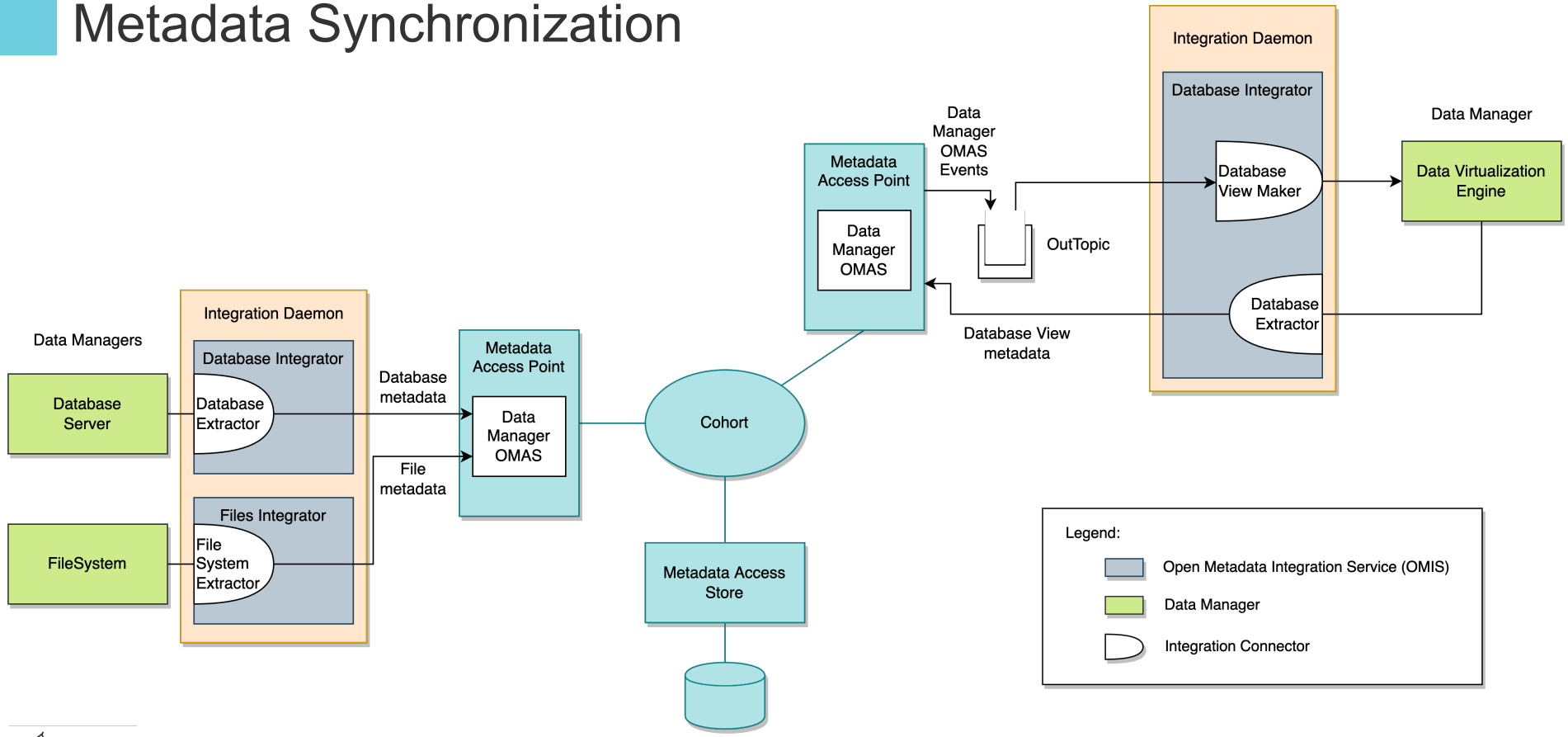




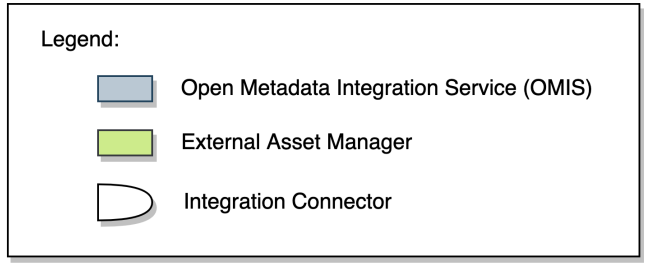
Metadata Distribution

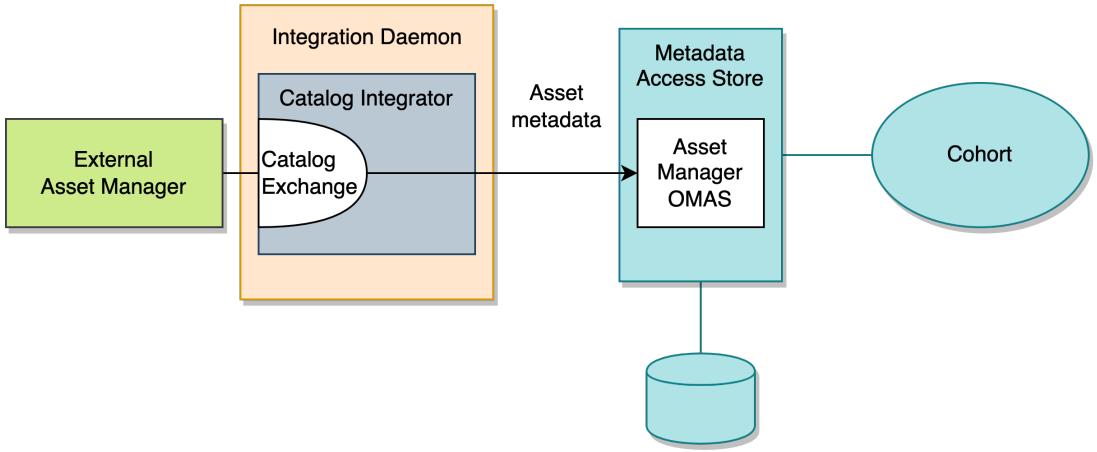


Metadata Synchronization

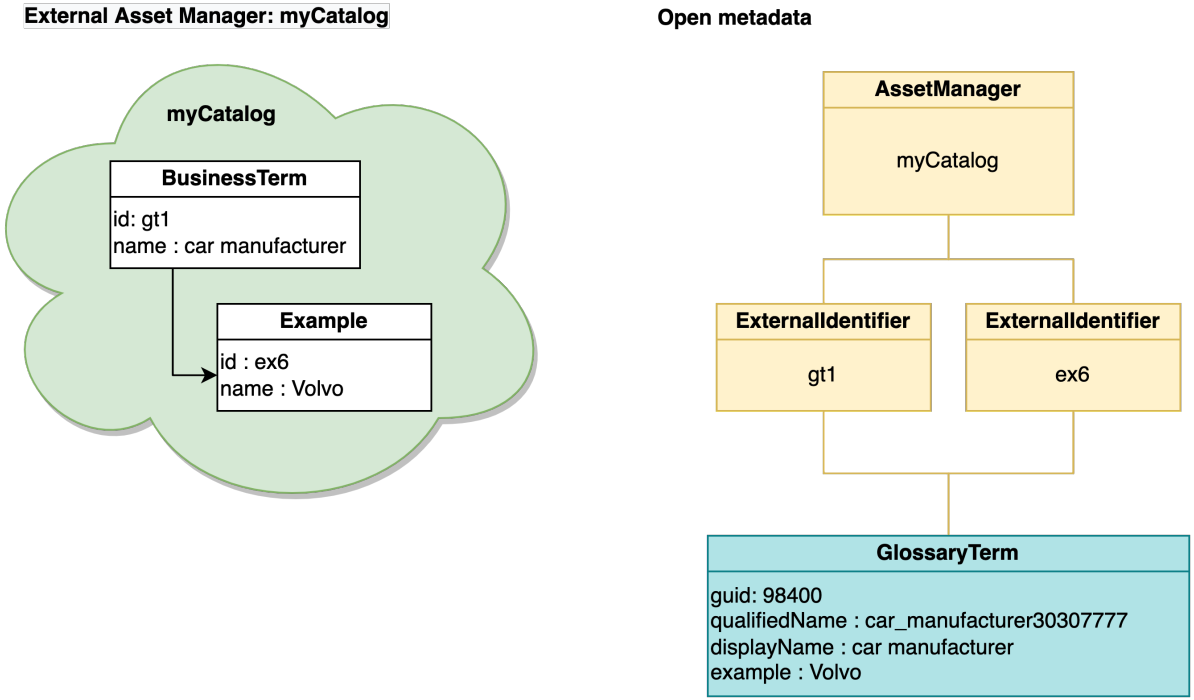


External Asset Manager

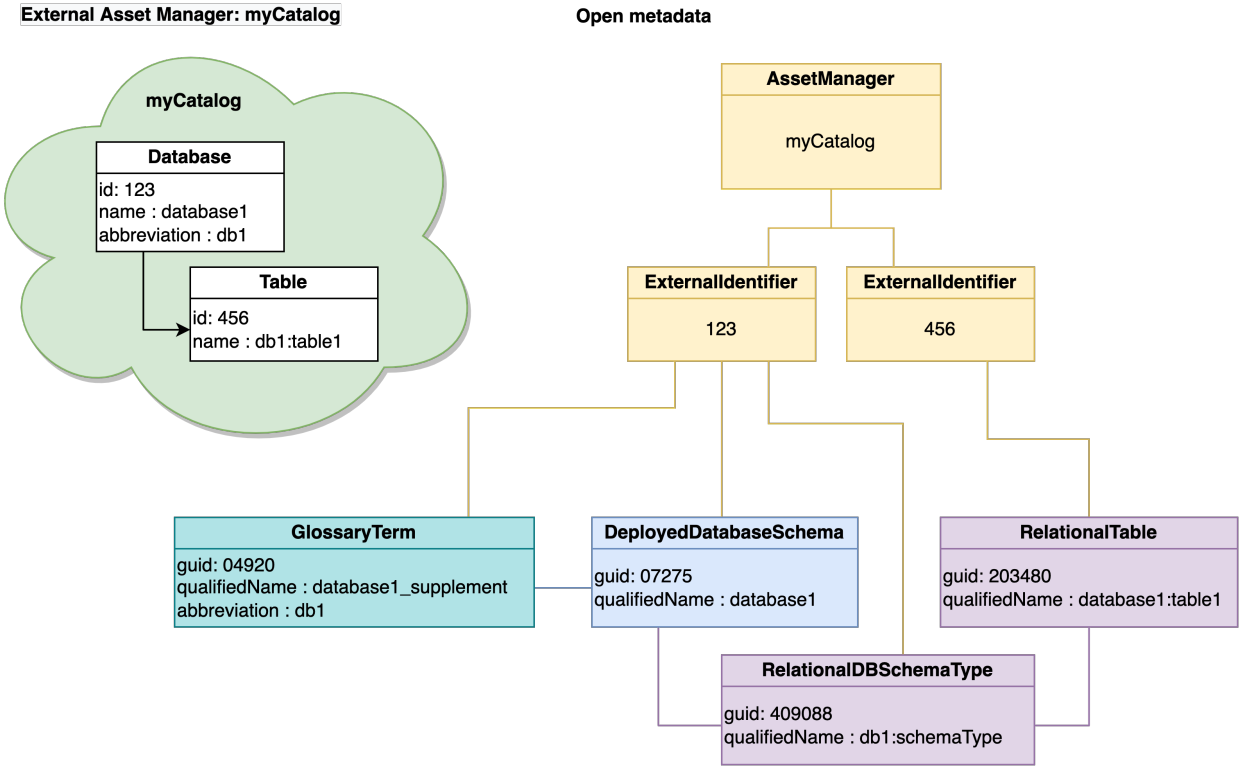




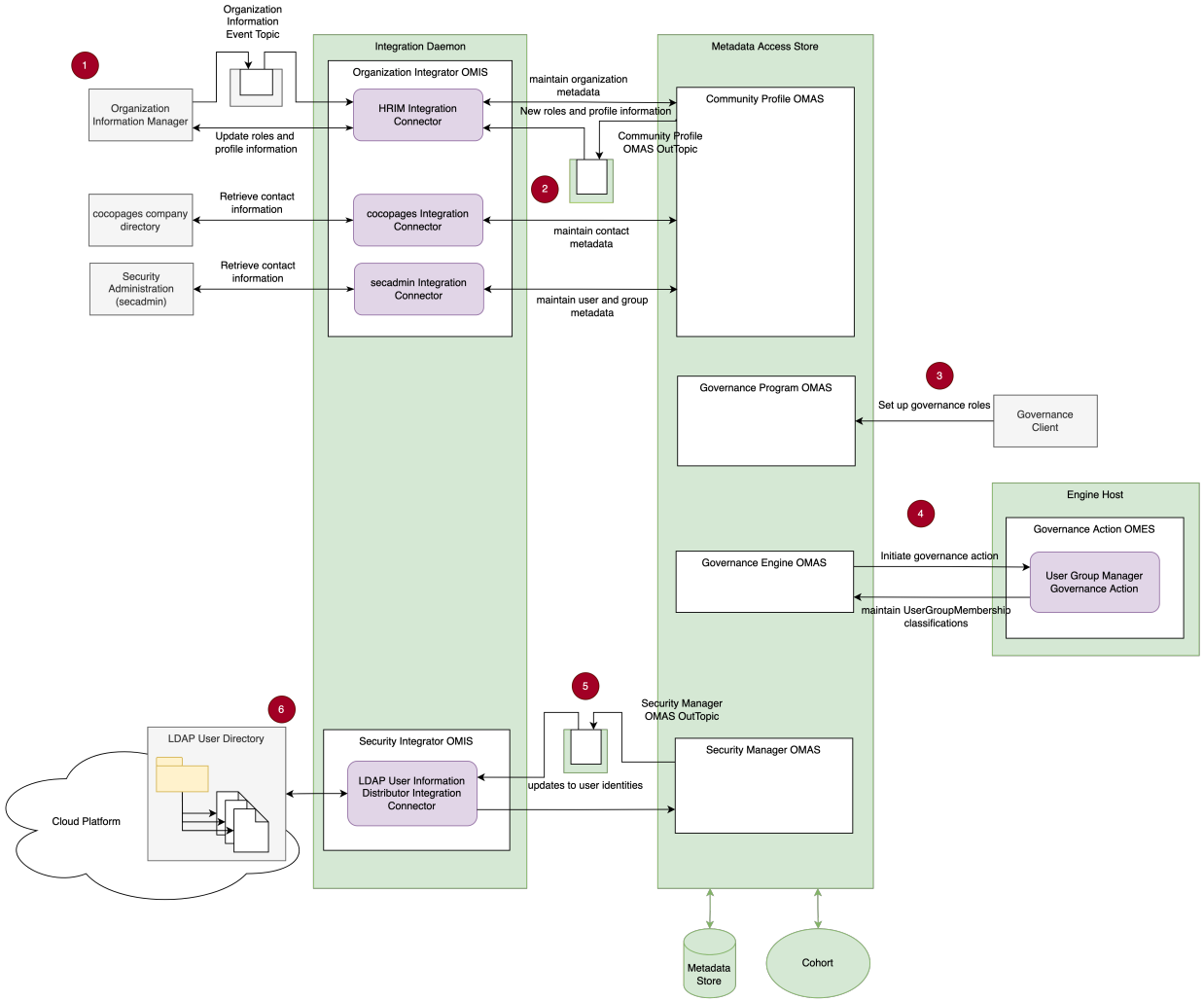
Many-to-One Mapping



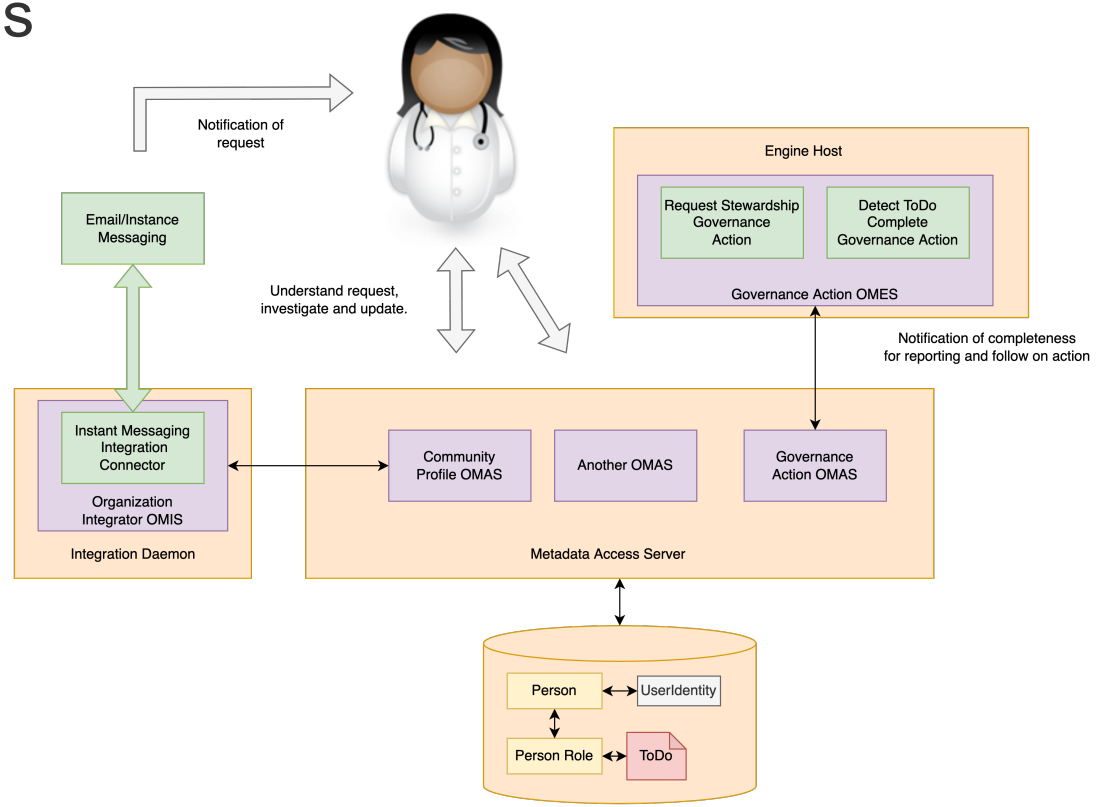
One-to-Many Mapping



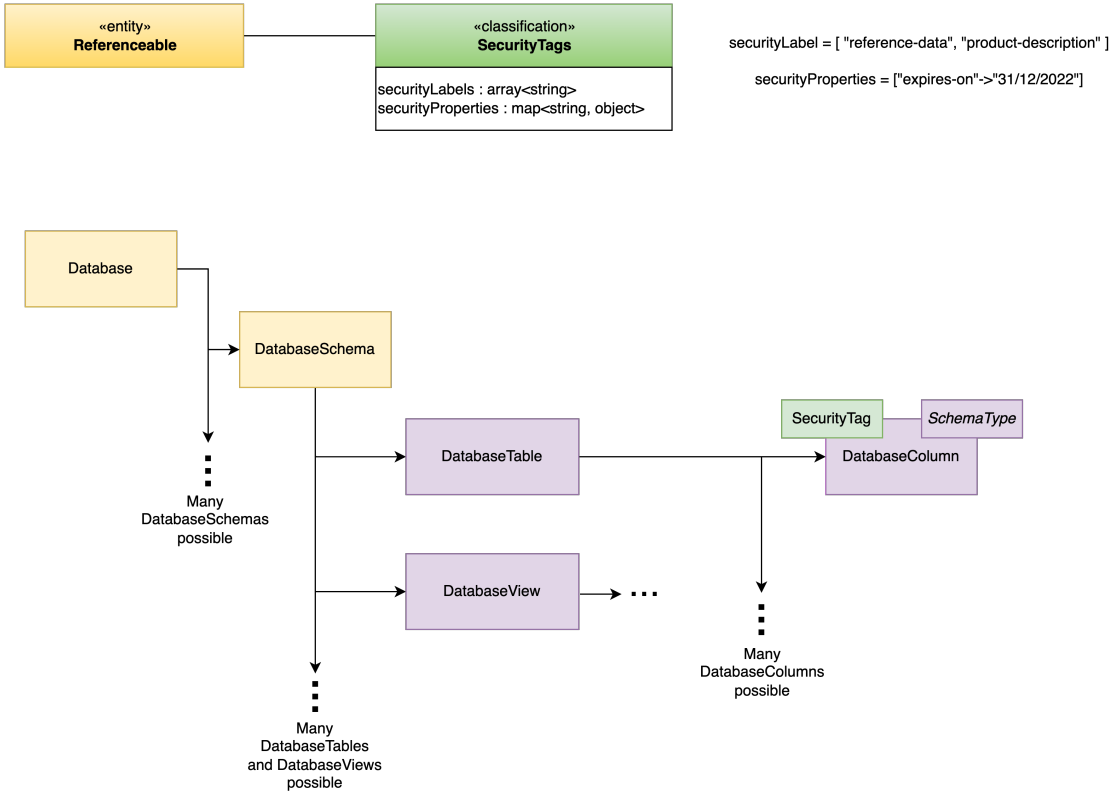
Onboarding an organization

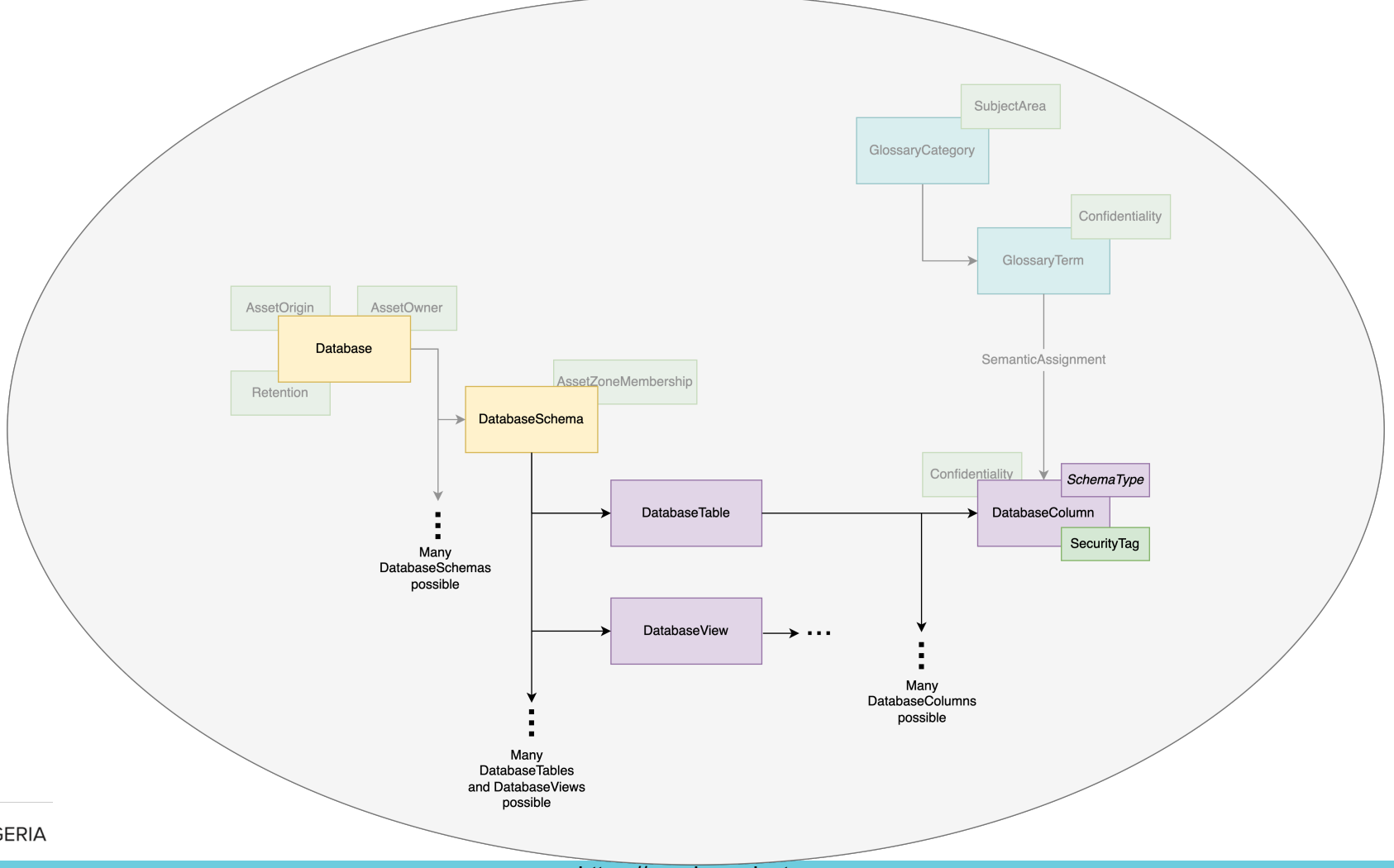


Steward Notifications

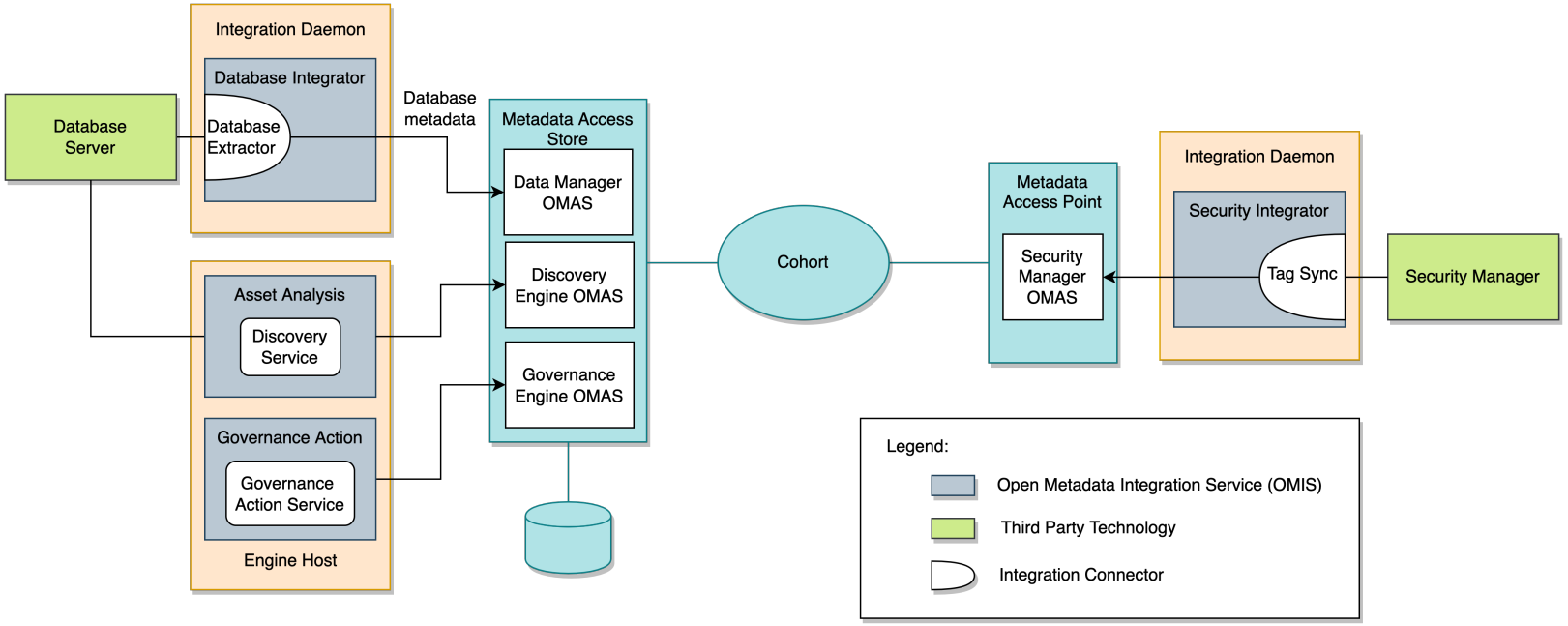


Pushing security tags to security managers

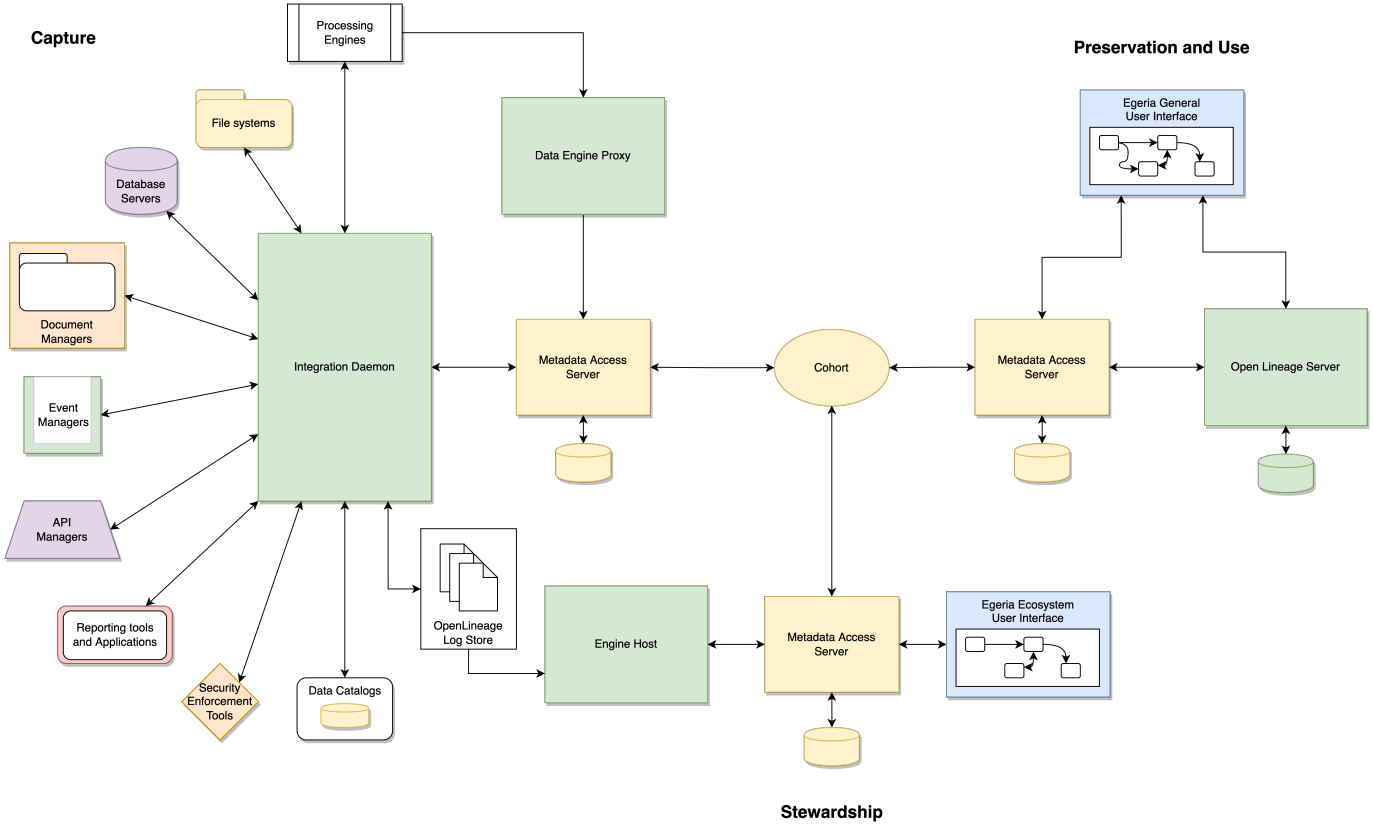



SubjectArea
GlossaryCategory
Confidentiality
GlossaryTerm
AssetOrigin
AssetOwner
Database
Retention
AssetZoneMembership
DatabaseSchema
SemanticAssignment
Confidentiality
SchemaType
DatabaseTable
DatabaseColumn
SecurityTag
Many
DatabaseSchemas
possible
DatabaseView
Many
DatabaseTables
and DatabaseViews
possible
Many
DatabaseColumns
possible

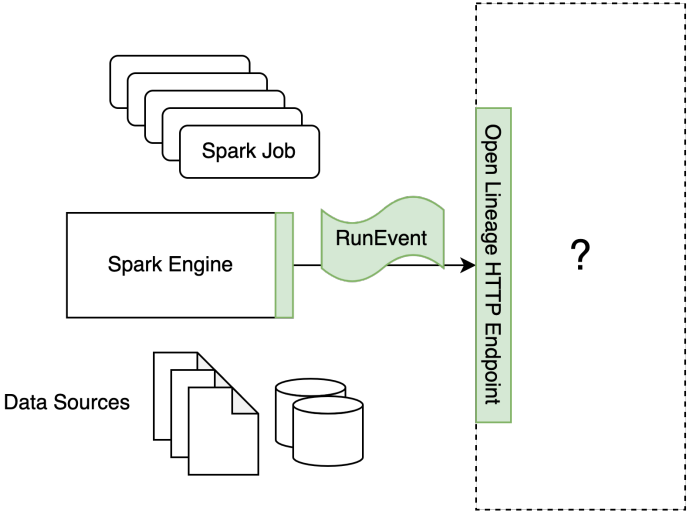
Pushing security tags to a security manager

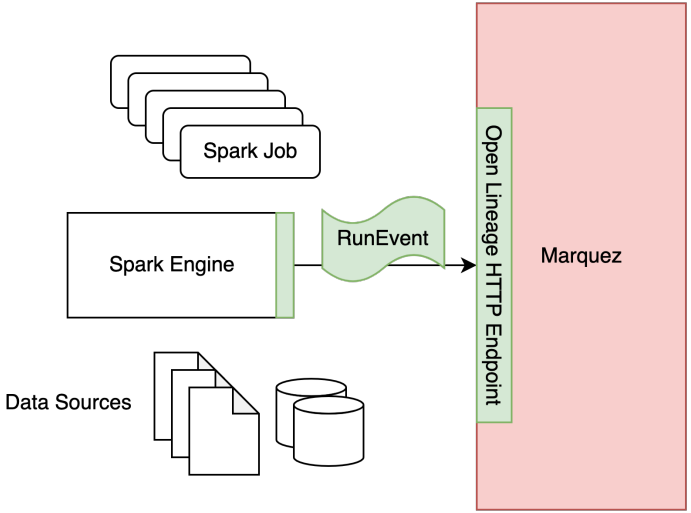


Lineage management

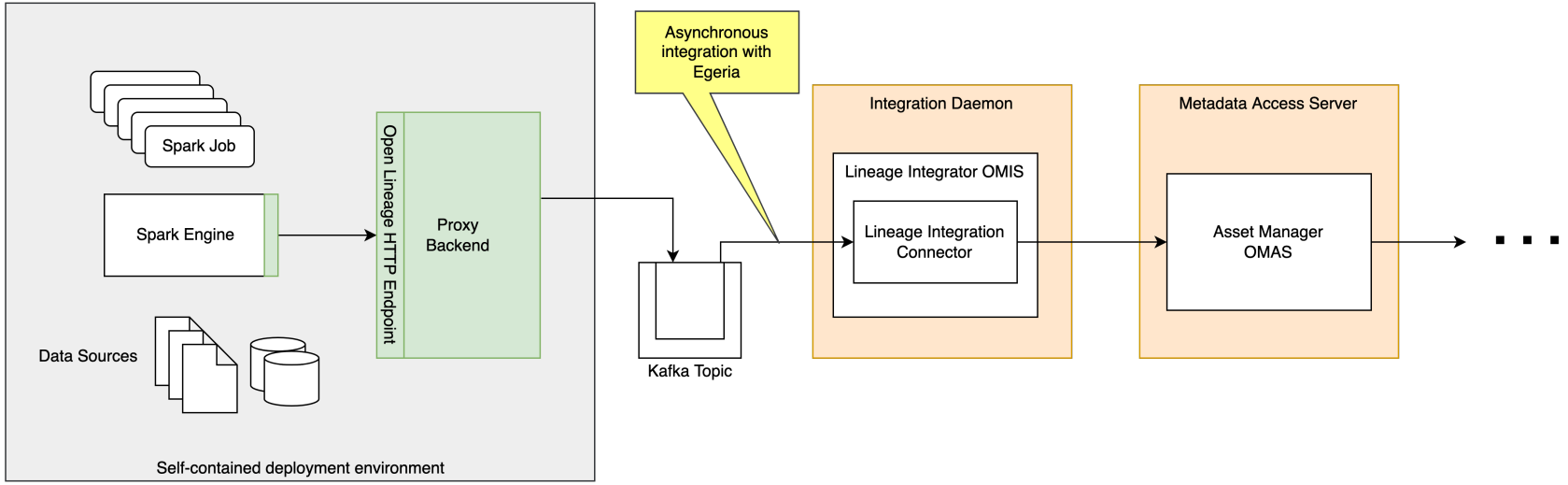


OpenLineage standard

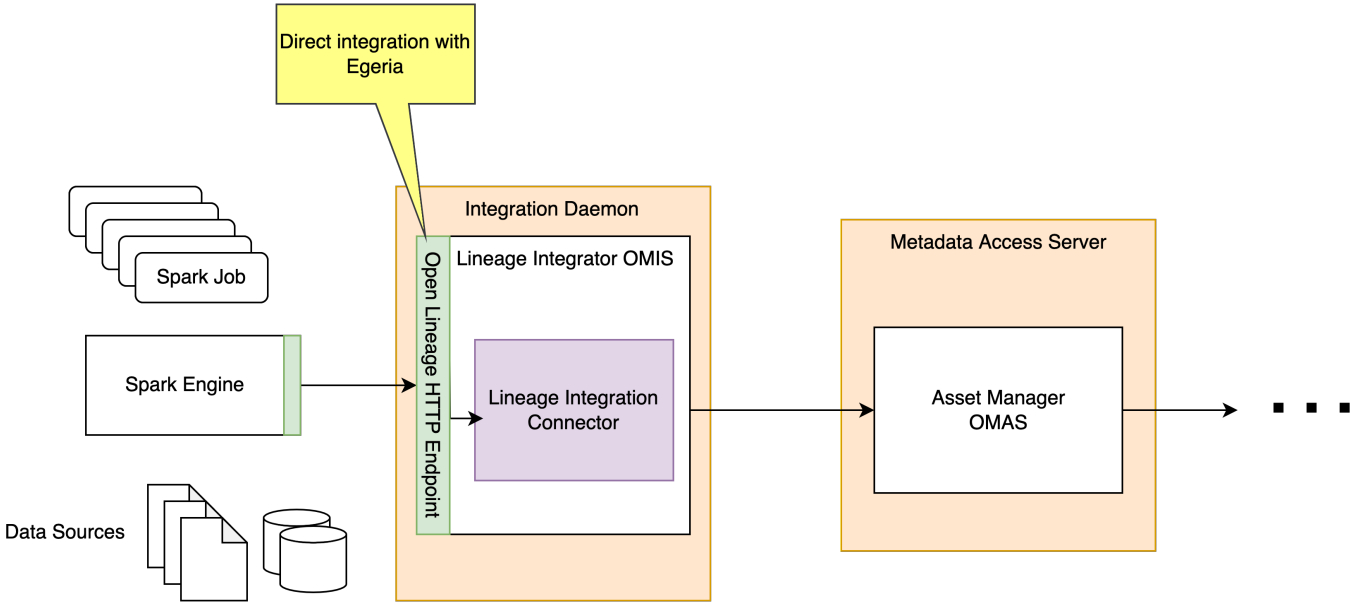




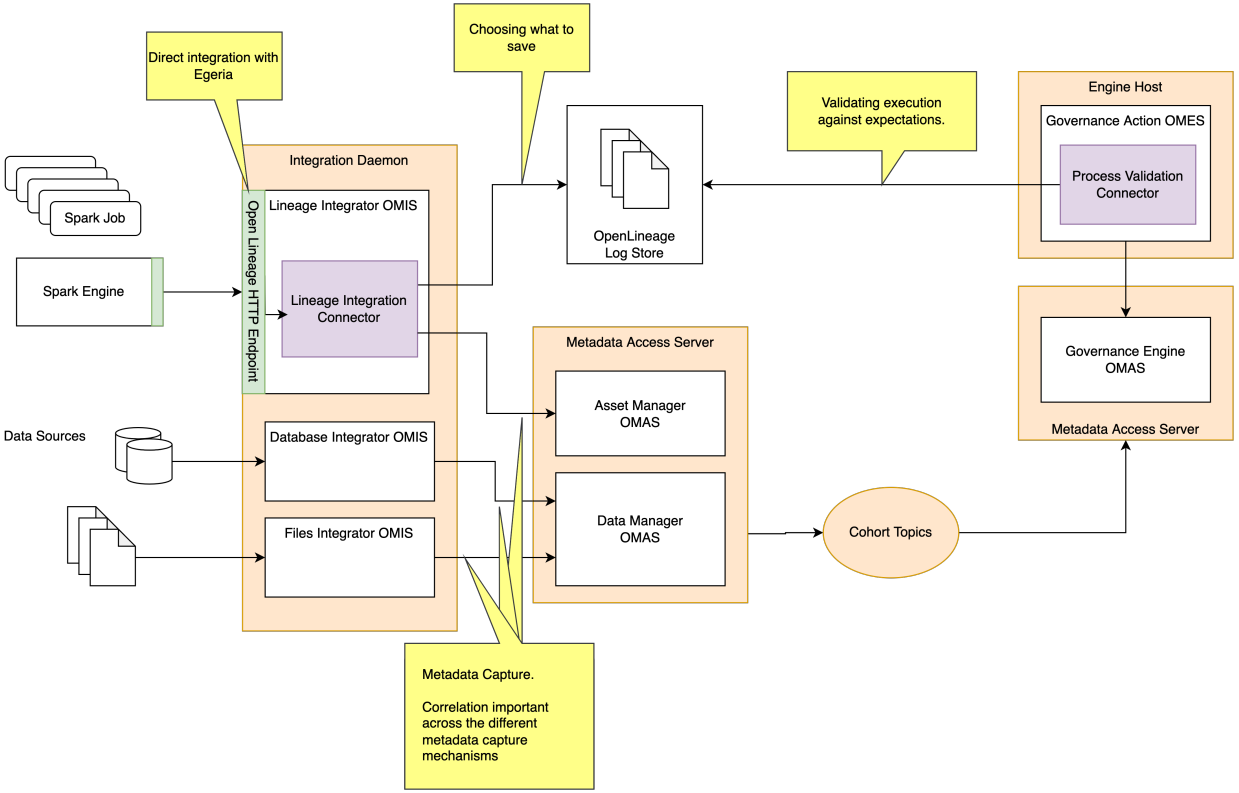
Receiving OpenLineage events



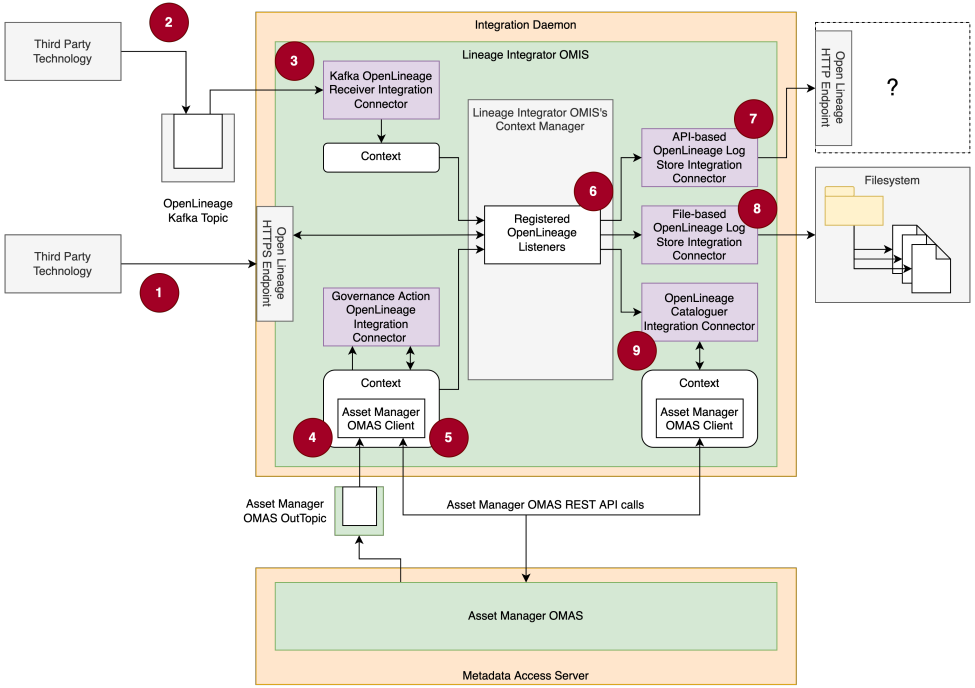
Direct receipt of OpenLineage events



Lineage Deployment



Inside the Lineage Integrator OMIS



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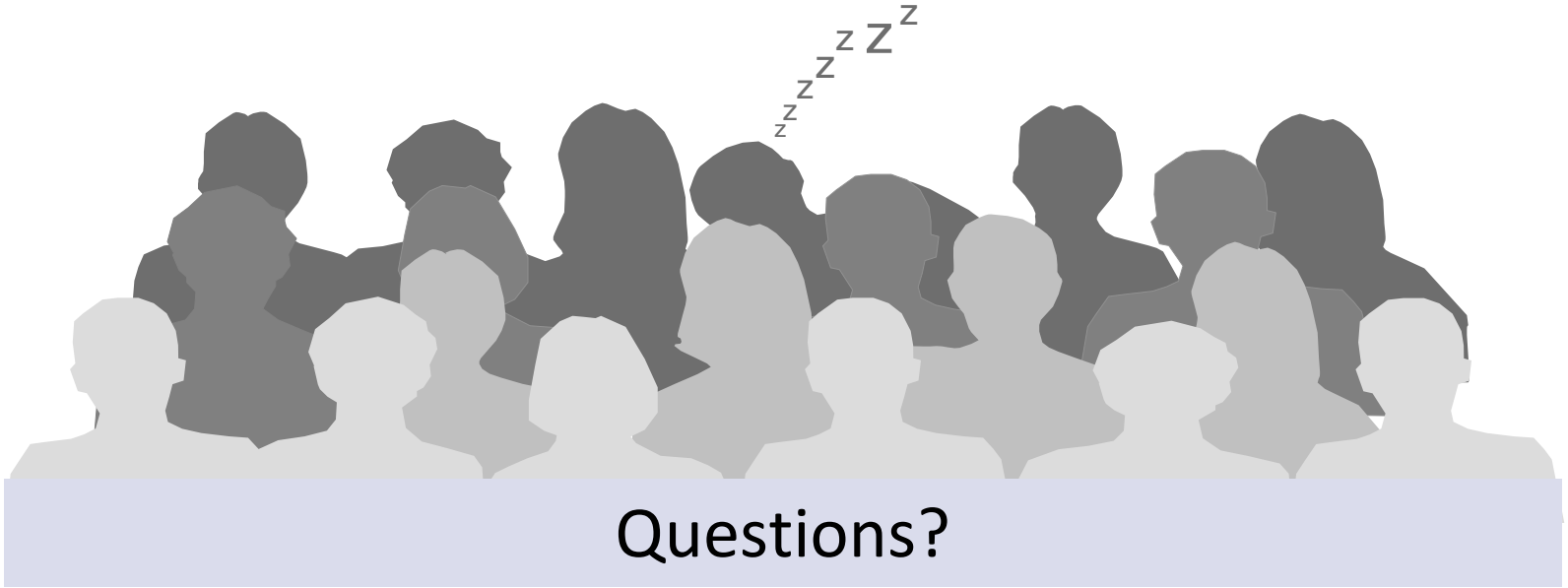
Open forum



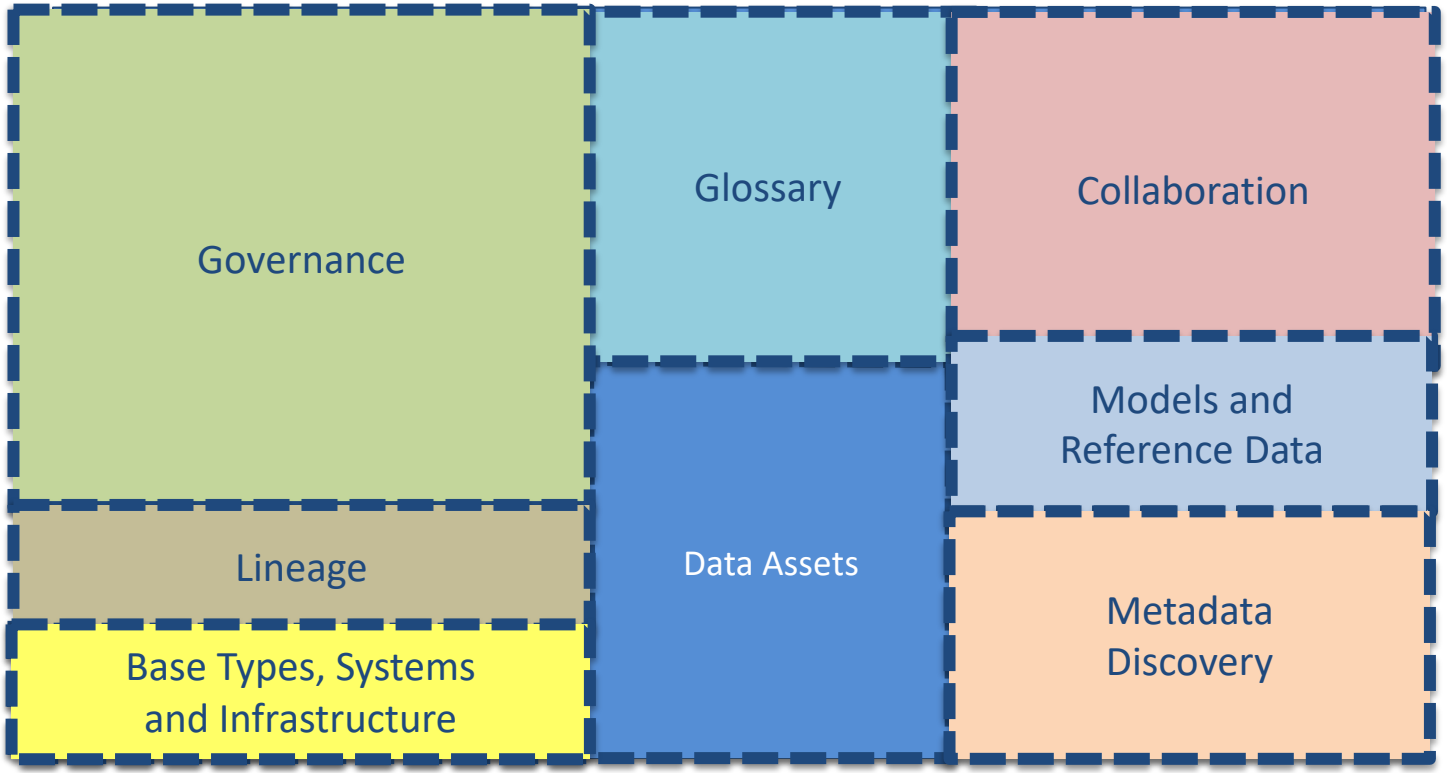
THANK YOU!

EGERIA

Achievements

- 700 linked open metadata types demonstrating how the knowledge from many tools can be linked together.
- Open metadata repository interface proven for table, graph and hierarchical DB stores.
- Enterprise queries and replication across heterogeneous technologies
- Conformance test suite and mark
- Automated configuration of data virtualization technology and security as new data sets are added to a data lake
- Suite of persona-based labs and tutorial using Jupyter Notebooks.
- Virtual graph of metadata maintained across distributed heterogenous metadata repositories.
- Frameworks, APIs and connectors for minimizing integration cost for different types of technologies
- Virtual repository explorer UI
- Instance based security
- Controlling visibility of assets through zones
- Scalable, secure platform configurable and customizable through connectors
- Purpose-based data access
- Metadata versioning and provenance
- Multi-tenant UI based on carbon
- W3C semantic standards pattern for data model exchange
- Automation of metadata acquisition through templates, daemons, discovery services and stewardship.
- Classification of assets
- Reference data management
- Multi-technology collaboration and feedback
- Multi-domain governance model
- Digital service lifecycle, from business design, development, devOps and use.
- Comprehensive open lineage services.
- Metadata deduplication

Scope of metadata covered



https://egeria.odpi.org/open-metadata-publication/website/open-metadata-types/

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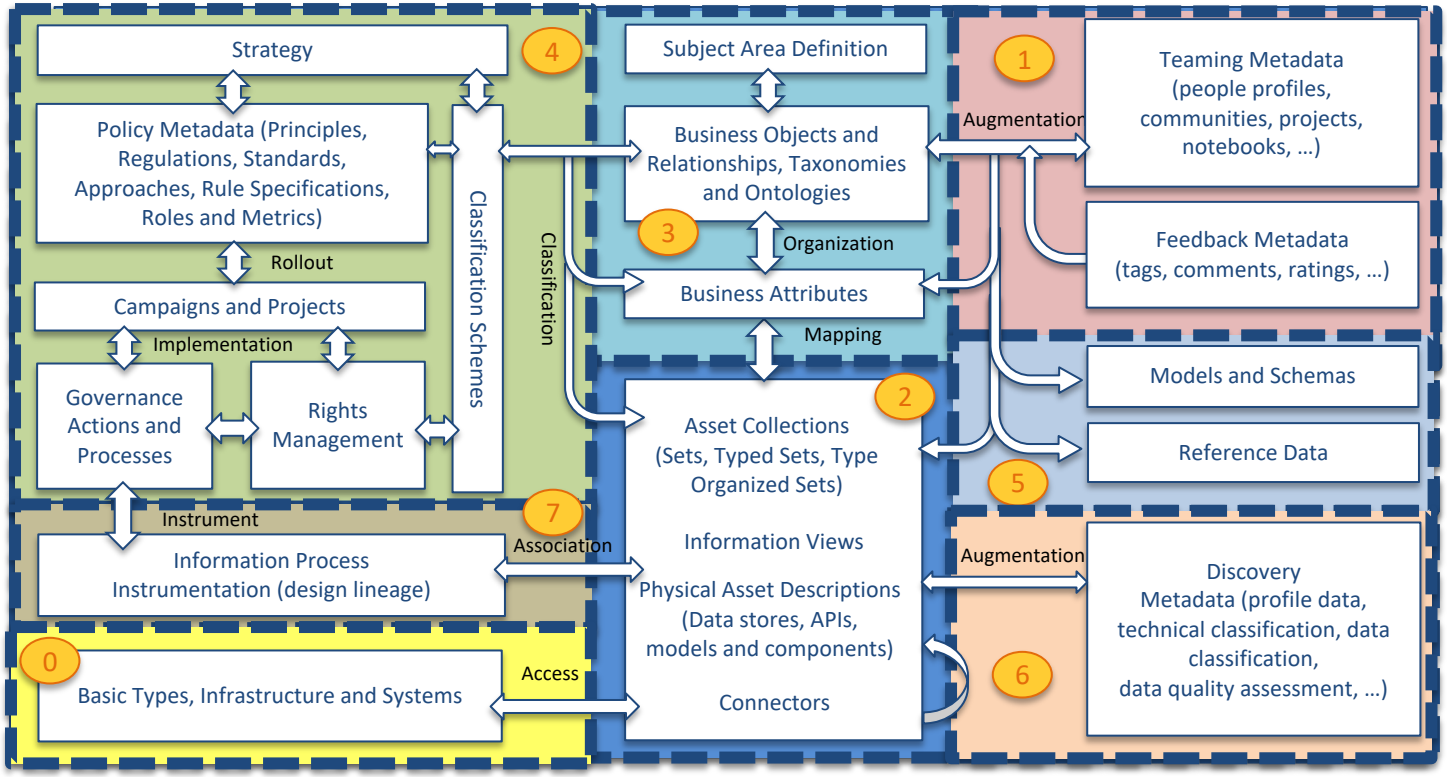

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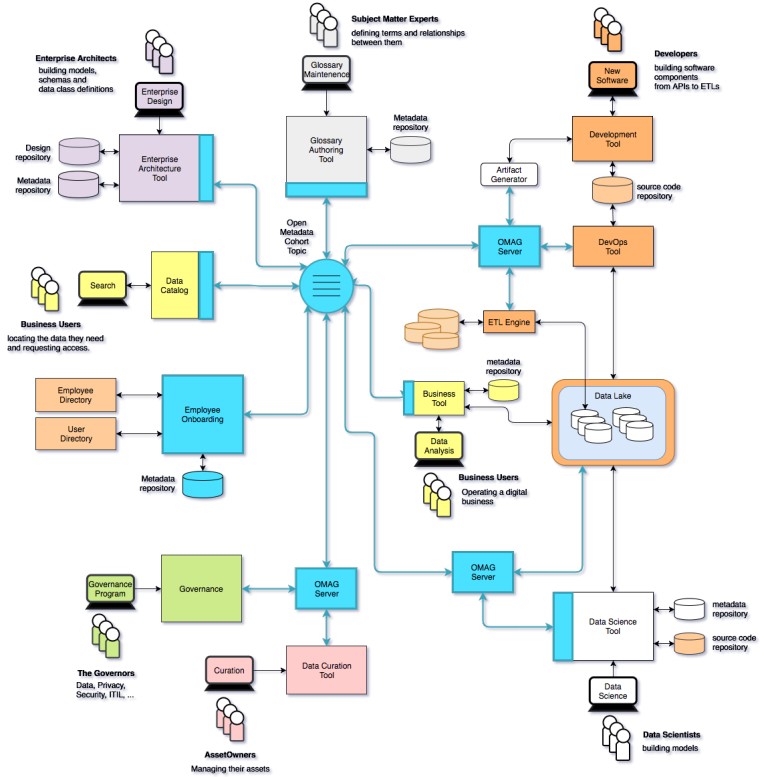
Using Egeria …

- Eases the cost of metadata integration through
  - Comprehensive standards and libraries.
  - Active vendor recruitment program.
- Provides direct support to many governance roles, filling the gaps between function offered through commercial tools.
- Provides best practices and content packs to accelerate an organization's journey to becoming data driven.